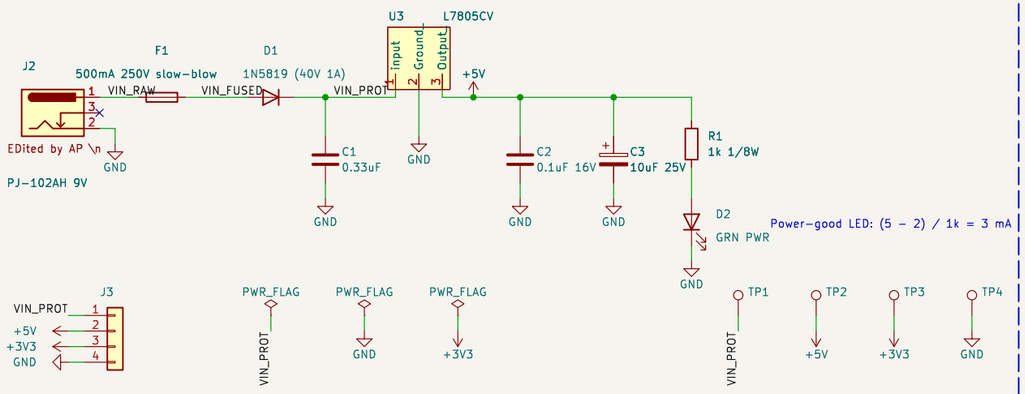
POWER INPUT 9–12 V → 5 V LINEAR REGULATOR

9 V wall adapter (Adafruit #63, 9–12 V OK) → 0.5 A slow-blow fuse (removable, TR5 socket) → 1N5819 reverse-polarity diode → L7805CV (TO-220) → +5V

Load est. < 100 mA → Pd = (12 − 5) x 0.1 = 0.7 W, OK without heatsink. LM7805 datasheet caps: 0.33 uF in, 0.1 uF out (+10 uF bulk)



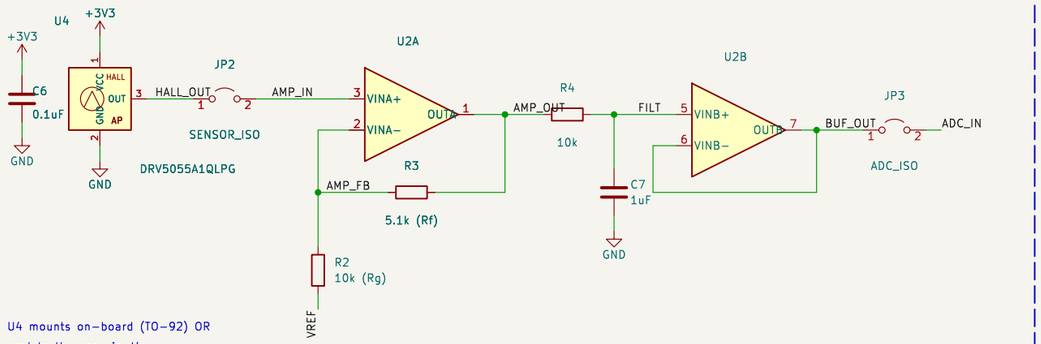
AMPLIFIED ANALOG SENSOR – HALL-EFFECT CAP / LID SENSOR (Board B)

DRV5055A1 (ratiometric) on +3V3: Vq = VCC/2 = 1.65 V, 60 mV/mT, +/−21 mT. Cap magnet north-pole-down → output drops when closed.

Stage 1 U2A gain about VREF: Vout = VREF + G(Vin − VREF), G = 1 + R3/R2 = 1 + 5.1k/10k = 1.51. VREF = 1.65 V from R8/R9, buffered by U2C

Stage 2 R4/C7 passive RC low-pass: fc = 1 / (2 pi R C) = 1 / (2 pi x 10k x 1 uF) = 15.9 Hz (cap motion < 5 Hz, removes noise)

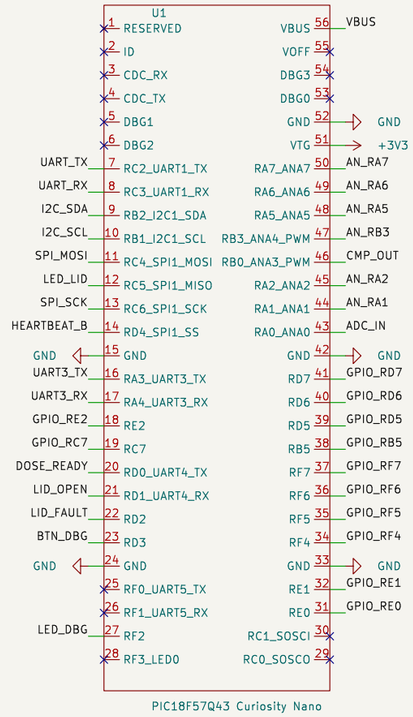
Stage 3 U2B unity-gain buffer → low-impedance drive into ADC RA0. All analog on +3V3 so nothing can exceed the PIC pin. U2D unused.

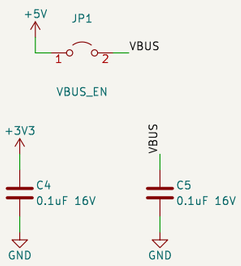


MICROCONTROLLER – PIC18F57Q43 CURIOSITY NANO (custom symbol _AP)

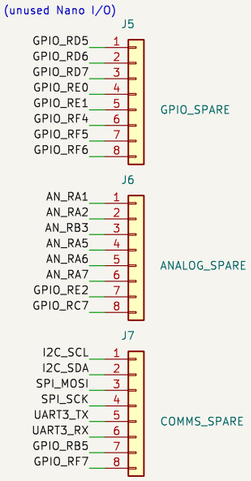
Nano runs at VTG = 3.3 V (default). +5V feeds VBUS through JP1 – REMOVE JP1 whenever the Nano USB cable is plugged in.

Hub link (J1 → Hub J2): RD0 DOSE_READY in, RD1 LID_OPEN / RD2 LID_FAULT / RD4 HEARTBEAT_B out, UART1 RC3 RX / RC2 TX.





Expansion headers (unused Nano I/O)

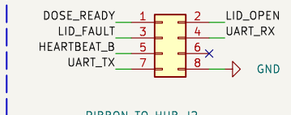


RIBBON J1 → HUB J2

8-pin ribbon, 3.3 V logic

1–5 digital, 6–7 analog, 8 GND

Pins 4 / 7 = UART (were spare)



1 DOSE_READY hub → B (RD0)
2 LID_OPEN B → hub (RD1)
3 LID_FAULT B → hub (RD2)
4 UART_RX hub → B (RC3)
5 HEARTBEAT_B B → hub (RD4)
6 spare (analog)
7 UART_TX B → hub (RC2)
8 GND

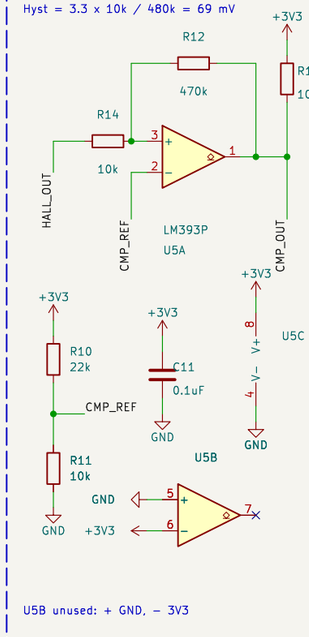
LID COMPARATOR (LM393)

Raw sensor vs 1.03 V threshold.

RB0 = 1 when cap OPEN (HALL > Vth)

Vth = 3.3 x 10k / 32k = 1.03 V

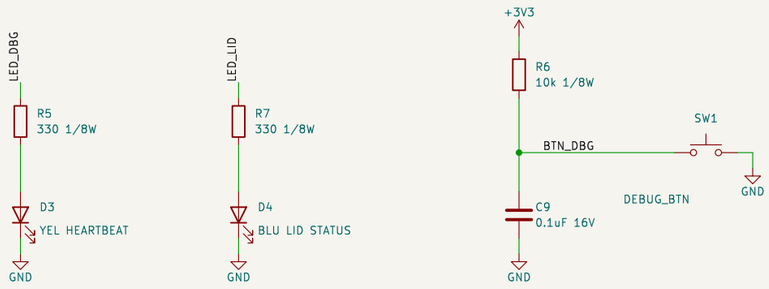
Hyst = 3.3 x 10k / 480k = 69 mV



U5B unused: + GND, − 3V3

STATUS / DEBUG LEDs + PUSH BUTTON

D3 = heartbeat on RF2 ("MCU is working"), D4 = lid status LED on RC5. I_LED = (3.3 − 2.0) / 330 = 4 mA. SW1 on RD3



R6 pull-up: idle = 3.3 V, pressed = 0 V. C9 debounce, tau = R6 x C9 = 1 ms

PIC18F57Q43 Curiosity Nano + L7805CV + DRV5055A1 Hall + MCP6004 amp/LPF/buffer + LM393 co

Arya Padiyar – Individual Subsystem Schematic – Rev 2 (matches Team 102 block diagram)

EGR 304 Embedded Systems Design – Team 102

Sheet: /

File: BoardB_HallSensor.kicad_sch

Title: Board B – Lid Sensing Subsystem (Hall Effect)

Size: A3	Date: 2026-10-19	**Rev: 2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1